MAE 6246 - Linear Systems and Control

Week 9: Controllability Decomposition and Pole Placement

Fall 2026

October 23, 2026

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Lecture: Friday, October 23, 2026

Main question: Which eigenvalues can state feedback change, and how do we choose a gain to put them where we want?

Scope: controllability decomposition; an introduction to feedback control; stabilizability; pole placement through coefficient matching and canonical coordinates; numerical verification and practice.

Prerequisites: Controllability rank and PBH tests; range spaces and bases; similarity transformations; characteristic polynomials.

1 From a subspace to a decomposition

1.1 Model and coordinate convention

We consider the continuous-time, real LTI system

$$\dot{x} = Ax + Bu, \qquad x \in \mathbb{R}^n, \quad u \in \mathbb{R}^m.$$

The state is assumed available for feedback. Observer design will address how to obtain it from measurements. Our convention throughout is

$$\boxed{x = Qz, \qquad z = Q^{-1}x, \qquad u = -Kx.}$$

A matrix is *Hurwitz* if every eigenvalue has strictly negative real part.

1.2 The controllable subspace and its invariance

The controllability matrix and controllable subspace are

$$\mathcal{C} = [B, AB, \ldots, A^{n-1}B], \qquad \mathcal{R} = \operatorname{im} \mathcal{C}, \qquad r = \dim \mathcal{R}.$$

Starting at zero, all states reachable by an input lie in $\mathcal{R}$. In continuous time, every vector in $\mathcal{R}$ can be reached over any specified positive time interval with an unconstrained input.

Invariance under A. Multiplying a generating column A^jB by A gives the next column. For the last one, Cayley–Hamilton expresses A^nB as a linear combination of $B, \ldots, A^{n-1}B$. Hence

$$A\mathcal{R} \subseteq \mathcal{R}, \qquad \operatorname{im} B \subseteq \mathcal{R}.$$

1.3 Constructing the block form

Choose a basis $Q_c \in \mathbb{R}^{n\times r}$ for $\mathcal{R}$ and complete it to an invertible matrix $Q = [Q_c \; Q_u]$. Because AQ_c and B lie in $\operatorname{im} Q_c$,

$$AQ_c = Q_c A_c, \qquad B = Q_c B_c.$$

Therefore, in the coordinates $z = [z_c^T, z_u^T]^T$,

$$\boxed{\dot{z} = \begin{bmatrix} A_c & A_{12} \\ 0 & A_u \end{bmatrix} z + \begin{bmatrix} B_c \\ 0 \end{bmatrix} u.}$$

Here $A_c \in \mathbb{R}^{r\times r}$ and $A_u \in \mathbb{R}^{(n-r)\times(n-r)}$. The lower-left zero follows from invariance, and the lower input block is zero because $\operatorname{im} B \subseteq \mathcal{R}$.

2 What the decomposition says

2.1 The controllable and uncontrollable blocks

The transformed matrices are

$$\bar{A} = Q^{-1}AQ = \begin{bmatrix} A_c & A_{12} \\ 0 & A_u \end{bmatrix}, \qquad \bar{B} = Q^{-1}B = \begin{bmatrix} B_c \\ 0 \end{bmatrix}.$$

Moreover,

$$Q^{-1}\mathcal{C} = \begin{bmatrix} B_c & A_cB_c & \cdots & A_c^{n-1}B_c \\ 0 & 0 & \cdots & 0 \end{bmatrix}.$$

Its rank is r. By Cayley–Hamilton for A_c, the first r block columns already span the upper space. Thus (A_c, B_c) is controllable.

Interpretation. The coordinate z_u obeys $\dot{z}_u = A_u z_u$ independently of the input and of z_c. It can drive z_c through A_{12}. The complement $\operatorname{im} Q_u$ need not itself be invariant; the decomposition need not be block diagonal. The eigenvalues of A_u are the uncontrollable eigenvalues, including their multiplicities. If $y = Cx + Du$, then $y = [CQ_c\ CQ_u]z + Du$; the decomposition makes no claim about observability.

2.2 Example: constructing Q by hand

Consider the three-state system:

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \qquad \mathcal{C} = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix}.$$

Since $\operatorname{rank}\mathcal{C} = 2$, choose $q_1 = B$, $q_2 = AB$, and $q_3 = e_1$:

$$Q = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \qquad Q^{-1} = \begin{bmatrix} 0 & 1 & -1 \\ 0 & 0 & 1 \\ 1 & 0 & -1 \end{bmatrix}.$$

Direct multiplication gives

$$\boxed{\bar{A} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad \bar{B} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.}$$

Thus $A_c = \begin{bmatrix} 0 & -1 \\ 1 & 2 \end{bmatrix}$, $B_c = [1, 0]^T$, $A_u = [1]$, and $A_{12} = 0$. Check

$$[B_c, A_cB_c] = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

The uncontrollable coordinate is $z_u = x_1 - x_3$, so $\dot{z}_u = z_u$. The reachable plane is $x_1 = x_3$. Although all three open-loop eigenvalues are $+1$, only two belong to the controllable block.

3 From system analysis to feedback control

3.1 What is the controller supposed to do?

So far, we have analyzed a plant: given A, B, an initial state, and an input, what motion results? In control design we reverse the question: *how should we choose the input so that the motion has a desired behavior?* Typical objectives include stabilizing an equilibrium, reducing oscillations, following a reference, and rejecting disturbances.

Here we begin with **regulation**: drive the state toward the desired equilibrium and keep it there. We write that equilibrium as the origin. If a constant operating point satisfies $Ax_\star + Bu_\star = 0$, define $\delta x = x - x_\star$ and $\delta u = u - u_\star$. Then $\delta\dot{x} = A\delta x + B\delta u$. Thus regulating the origin also describes returning to a nonzero operating point after a deviation. For a nonlinear plant, these equations can instead be its local linearization.

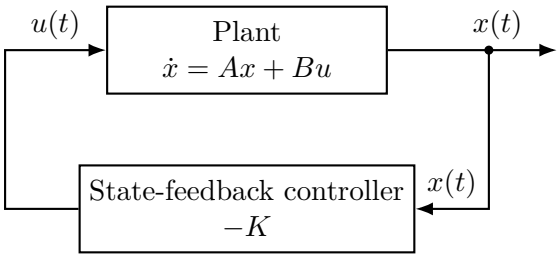
3.2 Open-loop input versus feedback

An open-loop input is prescribed without using the current state. For example, a precomputed force can move a mass along a nominal trajectory. If the initial condition differs from the planned value, that same input need not correct the resulting error.

A feedback controller uses current information about the plant to choose its input. With the full state available, the simplest time-invariant linear choice is

$$\boxed{u(t) = -Kx(t), \qquad K \in \mathbb{R}^{m\times n}.}$$

This is *static full-state feedback*: static means that the controller has no internal dynamic state. Each actuator input is a weighted sum of state components. In deviation coordinates, the physical command is $u = u_\star - K(x - x_\star)$.



State feedback for regulation about the origin.

The plant's state $x(t)$ travels around the feedback path to the gain $-K$. The controller generates $u(t) = -Kx(t)$ and applies it to the plant. The minus sign is included in the gain block, so no additional subtraction is needed. The signal shown is the full state, rather than a measured output $y = Cx$.

3.3 Why this particular form?

A linear law preserves the linear model and makes the design question explicit. Substitution closes the loop:

$$\dot{x} = Ax + B \underbrace{(-Kx)}_{\text{controller output}} = \boxed{(A - BK)x}, \qquad x(t) = e^{(A-BK)t}x(0).$$

We can therefore choose K to change the natural motion. If $A - BK$ is Hurwitz, every initial state converges to zero in the ideal model. Pole placement chooses its eigenvalues to shape that motion.

The minus sign is a **convention**, not a restriction: $u = Lx$ is the same family of laws with $L = -K$. Entries of K need not be positive, and writing a minus sign does not by itself guarantee stability. The matrix $A - BK$ must be checked.

3.4 Physical example: a restoring force and damping

For $m\ddot{q} = u$, choose $x = [q, \dot{q}]^T$ and

$$u = -k_p q - k_v \dot{q} \quad \Longrightarrow \quad m\ddot{q} + k_v \dot{q} + k_p q = 0.$$

Positive k_p acts like a restoring spring and positive k_v like a damper. Together they give asymptotic regulation; either term alone is insufficient for this mass. This is state feedback with $K = [k_p, k_v]$.

Scope. We assume the full state can be measured or supplied by an estimator and first design regulation without actuator limits. A tracking command or disturbance model requires additional design. The next question is whether the input can influence every mode: the controllability decomposition tells us exactly which poles feedback can move.

4 Feedback, fixed eigenvalues, and stabilizability

4.1 Which eigenvalues can feedback move?

Apply full-state feedback $u = -Kx$. In transformed coordinates,

$$u = -KQz = -\bar{K}z, \qquad \bar{K} = KQ = [K_c \; K_u].$$

Consequently,

$$\boxed{\bar{A} - \bar{B}\bar{K} = \begin{bmatrix} A_c - B_cK_c & A_{12} - B_cK_u \\ 0 & A_u \end{bmatrix}.}$$

The determinant of a block triangular matrix is the product of its diagonal-block determinants:

$$\det(sI - A + BK) = \det(sI_r - A_c + B_cK_c)\det(sI_{n-r} - A_u).$$

Therefore the closed-loop eigenvalues consist of the assignable eigenvalues of $A_c - B_cK_c$ together with the fixed eigenvalues of A_u.

Neither K_c nor K_u can move an uncontrollable eigenvalue. Taking $K_u = 0$ is sufficient for eigenvalue assignment; the original-coordinate gain is

$$\boxed{K = [K_c \; 0]Q^{-1}.}$$

4.2 Stabilizability

The pair (A, B) is *stabilizable* if some real gain K makes $A - BK$ Hurwitz. The decomposition gives

$$\boxed{(A, B) \text{ stabilizable} \iff A_u \text{ Hurwitz.}}$$

Necessity follows because the eigenvalues of A_u cannot move. For sufficiency, place the controllable-block eigenvalues in the open left half-plane; both diagonal blocks are then Hurwitz, so the full matrix is Hurwitz even when the upper-right block is nonzero.

The equivalent PBH condition is

$$\boxed{\operatorname{rank}[\lambda I - A, \; B] = n \quad \text{for every } \lambda \in \mathbb{C} \text{ with } \operatorname{Re}\lambda \geq 0.}$$

Only eigenvalues of A in that region need checking. Controllability checks *all* modes; stabilizability permits uncontrollable modes if they decay by themselves.

Return to the example. Since $A_u = 1$, the three-state system cannot be stabilized by any state feedback. In contrast, $A = \operatorname{diag}(1, -2)$ with $B = [1, 0]^T$ is uncontrollable but stabilizable: $K = [4, 0]$ gives $A - BK = \operatorname{diag}(-3, -2)$.

Boundary matters. An uncontrollable pole at zero or on the imaginary axis prevents asymptotic stabilization. Merely avoiding positive real parts is insufficient.

5 Pole placement for a controllable pair

5.1 The pole-placement theorem

Pole-placement theorem. For real A, B, the pair (A, B) is controllable if and only if every real monic polynomial of degree n can be realized as $\det(sI - A + BK)$ by some real gain K. Equivalently, any desired multiset of n eigenvalues can be assigned, provided complex eigenvalues occur in conjugate pairs.

The theorem assigns eigenvalues and their algebraic multiplicities. It does not promise an arbitrary eigenvector structure, good conditioning, small control effort, or a particular tracking response. We develop the single-input construction below; the theorem also holds for multiple inputs.

5.2 Example: coefficient matching

Consider

$$\dot{x} = \begin{bmatrix} 0 & 1 \\ -2 & 0 \end{bmatrix} x + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u, \qquad u = -[k_1, k_2]x.$$

The controllability matrix is $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ and has rank two. The closed-loop matrix and characteristic polynomial are

$$A - BK = \begin{bmatrix} 0 & 1 \\ -2 - k_1 & -k_2 \end{bmatrix}, \qquad \det(sI - A + BK) = s^2 + k_2 s + (2 + k_1).$$

For desired poles $-2, -4$,

$$p_d(s) = (s + 2)(s + 4) = s^2 + 6s + 8.$$

Matching coefficients gives $k_2 = 6$ and $2 + k_1 = 8$, hence

$$\boxed{K = [6, 6], \qquad A - BK = \begin{bmatrix} 0 & 1 \\ -8 & -6 \end{bmatrix}.}$$

Its characteristic polynomial is $(s+2)(s+4)$, as required. This is regulation: in the absence of disturbances, every state tends to zero.

5.3 Choosing poles

For a second-order target $s^2 + 2\zeta\omega_n s + \omega_n^2$, the damping ratio ζ and natural frequency ω_n provide a useful starting point. Underdamped poles are

$$s = -\zeta\omega_n \pm i\omega_n\sqrt{1 - \zeta^2}, \qquad 0 < \zeta < 1.$$

For the example, $k_1 = \omega_n^2 - 2$ and $k_2 = 2\zeta\omega_n$. Poles farther left generally demand larger gains and more actuator effort. Check trajectories and $u(t)$ for representative initial states; eigenvalues alone do not describe transient amplification or saturation.

Tracking is a separate design step. With $u = -Kx + Nr$, the gain K still determines the homogeneous dynamics, but N or integral action is needed to specify the reference response. Pole placement alone does not guarantee zero tracking error.

6 Why coefficient matching works: canonical coordinates

6.1 Choosing the gain in canonical coordinates

Suppose the system has one input and (A, B) is controllable. There is an invertible Q such that

$$\bar{A} = Q^{-1}AQ = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ -a_0 & -a_1 & -a_2 & \cdots & -a_{n-1} \end{bmatrix}, \qquad \bar{B} = Q^{-1}B = e_n.$$

Here $\det(sI - A) = s^n + a_{n-1}s^{n-1} + \cdots + a_1 s + a_0$. For the target

$$p_d(s) = \prod_{j=1}^{n}(s - \mu_j) = s^n + d_{n-1}s^{n-1} + \cdots + d_1 s + d_0,$$

choose

$$\boxed{\bar{K} = [d_0 - a_0,\ d_1 - a_1, \ldots, d_{n-1} - a_{n-1}].}$$

Then $\bar{A} - \bar{B}\bar{K}$ has last row $[-d_0, \ldots, -d_{n-1}]$, so its characteristic polynomial is exactly p_d. Since $u = -\bar{K}z = -\bar{K}Q^{-1}x$,

$$\boxed{K = \bar{K}Q^{-1}.}$$

This explains the theorem constructively for a single input.

6.2 How to construct the canonical transformation

Construct $\bar{A}, \bar{B}$ from the open-loop polynomial above and form

$$\mathcal{C} = [B, AB, \ldots, A^{n-1}B], \qquad \bar{\mathcal{C}} = [\bar{B}, \bar{A}\bar{B}, \ldots, \bar{A}^{n-1}\bar{B}].$$

Both are invertible and $\mathcal{C} = Q\bar{\mathcal{C}}$, so

$$\boxed{Q = \mathcal{C}\bar{\mathcal{C}}^{-1}.}$$

The identity $AQ = Q\bar{A}$ follows from Cayley–Hamilton and the matching characteristic polynomial. The decomposition matrix on page 3 and the canonical transformation here serve different purposes: the former isolates the reachable subspace; the latter puts a controllable single-input system in coefficient-matching form.

6.3 Optional: Ackermann's formula

An equivalent direct formula for a controllable single-input pair is

$$\boxed{K = e_n^T\mathcal{C}^{-1}p_d(A), \qquad p_d(A) = A^n + d_{n-1}A^{n-1} + \cdots + d_0 I.}$$

This is useful for hand calculations on small systems. For larger numerical problems, avoid explicit matrix inverses and high powers when possible; use a numerical pole-placement routine and verify the resulting eigenvalues. Nearly uncontrollable systems can require very large gains.

7 A complete design in noncanonical coordinates

7.1 A force-driven mass with reordered states

Consider a force-driven unit mass with position q and velocity v, but store the state as $x = [v, q]^T$:

$$A = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Choose $z = [q, v]^T$, so

$$Q = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad Q^{-1} = Q, \quad \bar{A} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad \bar{B} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

For poles $-2, -4$, the desired polynomial is again $s^2 + 6s + 8$, and $a_0 = a_1 = 0$. Thus

$$\bar{K} = [8, 6], \qquad K = \bar{K}Q^{-1} = [6, 8], \qquad u = -6v - 8q.$$

In the original order,

$$A - BK = \begin{bmatrix} -6 & -8 \\ 1 & 0 \end{bmatrix}, \qquad \det(sI - A + BK) = s^2 + 6s + 8.$$

Check the state order. Using $[8, 6]$ directly with $x = [v, q]^T$ would implement the wrong feedback.

7.2 Python verification and simulation

The following verifies the pole-placement example on page 7. It uses NumPy, SciPy, and Matplotlib.

```
import numpy as np
from scipy.signal import place_poles
from scipy.linalg import expm
import matplotlib.pyplot as plt

A = np.array([[0., 1.], [-2., 0.]])
B = np.array([[0.], [1.]])
Ctrb = np.hstack([B, A @ B])
assert np.linalg.matrix_rank(Ctrb) == 2
K = place_poles(A, B, [-2., -4.]).gain_matrix
Acl = A - B @ K
print("K =", K)                  # [[6. 6.]]
print("poles =", np.linalg.eigvals(Acl))  # -2, -4

t = np.linspace(0., 5., 501)
x0 = np.array([1., 0.])
X = np.array([expm(Acl * ti) @ x0 for ti in t])
U = -(X @ K.T).ravel()
fig, ax = plt.subplots(2, 1, sharex=True, figsize=(6, 4))
ax[0].plot(t, X[:, 0], label="x1")
ax[0].plot(t, X[:, 1], label="x2")
ax[0].legend(); ax[0].set_ylabel("state")
ax[1].plot(t, U); ax[1].set_ylabel("input u")
ax[1].set_xlabel("time [s]")
fig.tight_layout(); plt.show()
```

8 Practice and brief answers

8.1 Practice

1. In the three-state example on page 3, derive $\dot{z}_u = z_u$ directly from $z_u = x_1 - x_3$. Explain why no feedback can make all states converge to zero.
2. For that example, place the two controllable poles at $-2, -4$. Find K_c, take $K_u = 0$, and compute the gain K in the original coordinates. What is the third closed-loop pole?
3. Let $A = \begin{bmatrix} 1 & 3 \\ 0 & -2 \end{bmatrix}$ and $B = [1, 0]^T$. Construct its controllability decomposition and find a stabilizing gain. Does the coupling entry 3 prevent stabilization?
4. For the example on page 7, choose poles $-3 \pm 4i$. Compute K and check the characteristic polynomial.
5. Explain why a controllable real system cannot have a real feedback gain whose only two closed-loop poles are $-1 + i$ and -3.
6. In the Python example, compare poles $(-2, -4)$ with $(-10, -20)$ using the same initial condition. Compare the state response, peak input, and gain magnitude.

8.2 Brief answers

1. $\dot{x}_1 = x_1 + x_2$, $\dot{x}_3 = x_2 + x_3$, so $\dot{z}_u = x_1 - x_3 = z_u$. For $z_u(0) \neq 0$, this coordinate grows as e^t regardless of input.
2. With $K_c = [k_1, k_2]$,
$$A_c - B_c K_c = \begin{bmatrix} -k_1 & -1 - k_2 \\ 1 & 2 \end{bmatrix},$$
whose characteristic polynomial is $s^2 + (k_1 - 2)s + (1 + k_2 - 2k_1)$. Matching $s^2 + 6s + 8$ gives $K_c = [8, 23]$. Therefore
$$K = [8, 23, 0]Q^{-1} = [0, 8, 15], \qquad \operatorname{spec}(A - BK) = \{-2, -4, 1\}.$$
The controllable modes stabilize, but the full system remains unstable.
3. The system is already decomposed: $Q = I$, $A_c = 1$, $B_c = 1$, $A_{12} = 3$, $A_u = -2$. Take $K = [4, 0]$; the closed-loop poles are $-3, -2$. The triangular coupling can affect transients but does not change these eigenvalues.
4. $p_d(s) = (s + 3)^2 + 16 = s^2 + 6s + 25$, so $K = [23, 6]$.
5. A real matrix has nonreal eigenvalues in conjugate pairs. The prescribed list omits $-1 - i$ and cannot be the spectrum of a real 2×2 matrix.
6. The faster choice gives $K = [198, 30]$ instead of $[6, 6]$. For $x_0 = [1, 0]^T$, $u(0)$ changes from -6 to -198. Inspect the entire simulated input history when comparing peak effort.

Scope. This handout treats the controllability part of the Kalman decomposition; the full controllability–observability decomposition is deferred.